

An AI-based Atlantic cod detection in an Arctic fjord: Linking Atlantic water intrusions to the cod (*Gadus morhua*) incursions

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Abstract

The Arctic Ocean is undergoing unprecedented environmental transformation, characterized by accelerated warming, reduction in sea ice cover, and altered oceanic circulation. These shifts are facilitating the poleward migration of boreal species, a process termed 'borealization', which is fundamentally restructuring Arctic marine ecosystems. Kongsfjorden, a high-latitude glacial fjord in Svalbard, serves as a critical natural laboratory for observing these changes due to its position at the confluence of Arctic and Atlantic water masses. Historically dominated by the endemic polar cod (*Boreogadus saida*), the fjord is now witnessing an increasing presence of Atlantic cod (*Gadus morhua*). This study, conducted under the Indian Arctic Programme, deployed a novel, non-invasive, artificial intelligence (AI)-driven optical monitoring system to document the fish community. Our results reveal a direct correlation between abundance of Atlantic cod in the inner Kongsfjorden and an intense intrusion of warm, saline Atlantic Water (AW) in late summer 2024. In contrast, cod were nearly absent during the early summer 2023 with weaker AW influence. These findings provide direct, high-resolution evidence for the oceanographic mechanism facilitating borealization in Arctic fjords and validate a new technology for sustained, long-term ecosystem monitoring in a remote and rapidly changing polar environment. The observed ecological shift signals a potential regime change that could restructure Arctic trophic interactions.

Key words: Arctic marine ecosystem, Atlantic cod, Polar cod, borealization, Atlantic water intrusion, deep learning

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Introduction

The Arctic is undergoing a period of profound and accelerated environmental change, warming at a rate nearly four times the global average in a phenomenon known as Arctic Amplification (Rantanen *et al.* 2022). This intensified warming is not an isolated thermal anomaly but the primary driver of a series of interconnected physical and ecological transformations across the cryosphere and marine ecosystems (Ardyna and Arrigo 2020, Polyakov *et al.* 2005). The consequences of the warming, include a dramatic reduction in the extent and thickness of sea ice, increased freshwater input from melting glaciers and rivers, and significant alterations to ocean circulation patterns (Polyakov *et al.* 2012, Rantanen *et al.* 2022). These physical shifts are fundamentally altering the biogeochemical landscape and habitat suitability for marine life, triggering a large-scale reorganization of Arctic biodiversity (Huntington *et al.* 2020).

This ecosystem reorganization is broadly characterized by borealization, a process whereby Arctic ecosystems, historically defined by cold-adapted, endemic species, progressively acquire the characteristics of more temperate, boreal systems (Husson *et al.* 2024). Borealization of Arctic marine ecosystems refers to the progressive acquisition of biological and physical characteristics associated with temperate and sub-Arctic systems. In the marine realm of the European Arctic, this process is predominantly driven by Atlantification, the increasing influence of warm, saline Atlantic-origin water that penetrates northward (Asbjørnsen *et al.* 2020, Polyakov *et al.* 2017). This advection of oceanic heat weakens the vertical stratification of the water column, reduces sea ice formation, and creates thermal conditions favourable for the northward expansion of marine boreal species (Carmack *et al.* 2015, Polyakov *et al.* 2017). The ecological ramifications are extensive, manifesting as pole-

ward shifts in the distribution of organisms across all trophic levels, from phytoplankton and zooplankton to commercially important fish stocks and top predators (Pinsky *et al.* 2013, Poloczanska *et al.* 2016, Pantiukhin *et al.* 2024). These redistributions are not merely additive; they lead to novel species interactions that may restructure food webs consequently. As a result of that, it may lead to alter ecosystem functions, with profound implications for overall stability and resilience (Fossheim *et al.* 2015, Renaud *et al.* 2015, Descamps *et al.* 2026).

Situated on the west coast of Spitsbergen, Svalbard, Kongsfjorden serves as one of the internationally recognized research sites for monitoring the impacts of climate change on high-latitude marine ecosystems (Svendsen *et al.* 2002, Piskozub 2017, Bischof *et al.* 2019). Its unique oceanographic setting makes it an ideal natural laboratory for studying the mechanisms and consequences of Arctic warming. As an open fjord lacking a shallow sill at its entrance, Kongsfjorden is highly susceptible to direct and dynamic cross-shelf exchange with the West Spitsbergen Current, the northernmost extension of the North Atlantic Current (Hop *et al.* 2002, Cottier *et al.* 2005). This facilitates the advection of warm, saline Atlantic Water (AW) into the fjord, which interacts with cold, fresh Arctic Water (ArW) from the coastal current and substantial meltwater discharge from large tidewater glaciers at the head of the fjord (Svendsen *et al.* 2002, Payne and Roesler 2019).

This confluence of water masses creates steep environmental gradients along the fjord's main axis, from the glacially influenced, turbid inner basin to the marine-dominated, clearer outer basin (Hop *et al.* 2002). This compressed ecosystem structure allows for the direct observation of biological responses to varying physical drivers. In recent decades, long-term mon-

itoring has documented a clear trend of increasing AW influence, leading to warmer water temperature, reduced seasonal sea ice cover, and a fundamental shift in the hydrographic regime of the fjord (Cottier et al. 2005, Payne and Roesler 2019). This progressive Atlantification has transformed Kongsfjorden into a system that may foreshadow the future state of other Arctic coastal environments, making it a critical location for understanding pan-Arctic change (Bischof et al. 2019, Dischereit et al. 2024).

The historical fish community of Kongsfjorden, and indeed much of the high-Arctic shelf ecosystem, has been overwhelmingly dominated by a single key species: the polar cod, *Boreogadus saida* (Hop et al. 2002). This small gadoid is not merely an abundant resident but a central pillar of the Arctic marine food web (Welch et al. 1992). With its high lipid content, the polar cod serves as a crucial energetic conduit, transferring energy from lower trophic levels (zooplankton and ice-associated crustaceans) to a wide array of top predators, including seals, whales, and seabirds (Hop and Gjøsæter 2013, Thorsteinson and Love 2016). The life cycle of the species is intimately linked to sea ice, which provides critical habitat for spawning, nursery grounds for juveniles, and refuge from predation (Gradinger et al. 2005, David et al. 2016, Malizia et al. 2023). This cryo-pelagic association makes polar cod exceptionally vulnerable to the rapid decline in sea ice cover, which constitutes a direct loss of essential habitat (Florko et al. 2021, Geoffroy et al. 2023).

Concurrent with the degradation of polar cod habitat, the warming of west Svalbard fjords has facilitated the northward expansion of a formidable boreal competitor- the Atlantic cod, *Gadus morhua* (Drinkwater 2005). While historically only occasional visitors, recent years have seen a marked increase in the abundance of Atlantic cod in Kongsfjorden, with stud-

ies confirming the presence of spawning adults and the successful establishment of juvenile populations in the shallow waters of the fjord (Szcucka et al. 2017, Spoto-witz et al. 2022, Brand et al. 2023). Atlantic cod is a larger, more aggressive, and more generalist predator than polar cod, with a higher thermal optimum (Kunz et al. 2016). Its establishment introduces intense new pressures of predation and competition for resources, posing a significant threat to the persistence of the native polar cod population (Fey and Węslawski 2017, Szcucka et al. 2017). This ongoing species replacement is not merely a change in community composition but signals a potential ecological regime shift, with far-reaching consequences for the structure and function of the entire fjord ecosystem.

The unprecedented rate of change in Arctic marine ecosystems demands a paradigm shift in monitoring strategies (Bicknell et al. 2016, Baschek et al. 2017). Traditional methods, such as ship-based trawl surveys, while foundational and important, are often logistically challenging, expensive, destructive to habitats to an extent, and provide only coarse, episodic snapshots of ecosystem dynamics (Bicknell et al. 2016). To adequately capture the rapid, often event-driven processes like species invasions, there is a critical need for high-resolution, continuous, and non-invasive monitoring technologies (Baschek et al. 2017). Advanced optical techniques, particularly when coupled with the analytical power of artificial intelligence (AI) and deep learning exploiting artificial neural networks, represent a transformative approach to modern biomonitoring (Philip et al. 2023). These systems can autonomously collect and process vast quantities of visual data, enabling the detection, classification, and quantification of marine organisms with high precision and temporal frequency. This approach complements other emerging non-invasive methods, such as environmental DNA (eDNA) metabarcod-

ing, to provide a more holistic view of biodiversity dynamics (Lacoursière-Roussel *et al.* 2018).

The present study focuses on documenting the fish community in Kongsfjorden and capturing the mechanism of borealization in real-time. The central hypothesis is that the influx of Atlantic cod in Kongsfjorden is not a gradual, uniform process but is instead facilitated by event-driven intrusions of warm Atlantic Water. By deploying a custom-built, AI-driven under-

water camera system, the primary objective of this research was to quantify the spatio-temporal extent of Atlantic cod intrusion in Kongsfjorden and to directly correlate their presence with prevailing hydrographic conditions. This approach allows for a mechanistic understanding of how large-scale climate change manifests as acute, high-impact biological events at the local scale, providing crucial insights for predicting the future trajectory of Arctic ecosystems.

Materials and Methods

Study area and field campaign

The study was conducted in Kongsfjorden (79°N, 12°E), a glacial fjord on the western coast of Spitsbergen, Svalbard (Fig. 1). Kongsfjorden is approximately 20 km long, varies in width from 4 to 10 km, and has a maximum depth of around 400 m near its mouth, with no sill to obstruct water exchange with the adjacent continental shelf (Svendsen *et al.* 2002). The fjord is characterized by a distinct environmental gradient, with several tidewater glaciers at its head influencing the inner basin and a stronger marine influence in the outer regions (Hop *et al.* 2002).

Field campaigns were carried out as part of the long-term Indian Arctic Programme, which has been conducting systematic monitoring in Kongsfjorden since 2011 from the Indian research station 'Himadri' in Ny-Ålesund. Data for this study were collected during three months across two years- the first during early summer (May-June) of 2023 and the second during late summer/early autumn (September) of 2024. Sampling was conducted at multiple stations distributed along the main axis of the fjord to represent the gradient from the inner to the outer fjord (Fig. 1). The specific coordinates for each sampling station are provided in Table 1.

Station Code	Latitude	Longitude
KG 1	78° 55.7134 N	11° 55.9985 E
KG 2	78° 54.191 N	12° 13.408 E
KG 3	78° 51.861 N	12° 35.028 E
KG 4	79° 02.750 N	11° 17.020 E
KG 5	79° 05.237 N	11° 18.039 E
KG 6	79° 04.094 N	11° 29.933 E
KG 7	79° 04.880 N	11° 40.657 E

Table 1. Coordinates of Sampling Stations in Kongsfjorden.

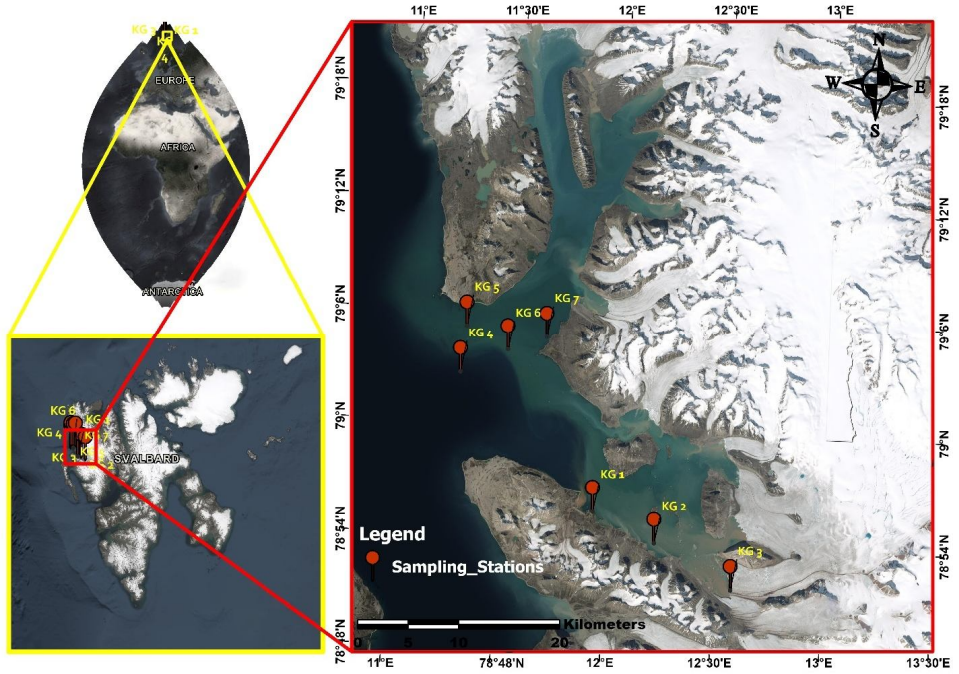


Fig 1. Map of the study area Kongsfjorden in Svalbard, marked with the sampling locations.

Hydrography data acquisition and water mass identification

Vertical profiles of physical properties of the Kongsfjorden water column were obtained at each station using a Sea-Bird Scientific SBE 19plus V2 Conductivity-Temperature-Depth (CTD) profiler. The instrument has an accuracy of $\pm 0.005^\circ\text{C}$ for temperature and ± 0.0005 S/m for conductivity used to derive salinity. Raw CTD data were processed following standard Seabird-recommended protocols, which included corrections for sensor lag, alignment, and bin-averaging to 1 m depth intervals to produce high-resolution profiles of temperature and salinity (David and

Krishnan 2017).

The processed temperature and salinity data were then used to classify the water masses present in the fjord during each sampling period. Water mass definitions were specific to Kongsfjorden and the West Spitsbergen Shelf (Cottier et al. 2005, Saghravani et al. 2024). This classification allowed for a quantitative assessment of the relative influence of different water sources, particularly the intrusion of Atlantic-origin waters. The specific criteria used for classification are detailed in Table 2.

Water Mass*	Abbreviation	Temperature (°C)	Salinity (PSU)	Characteristics and Origin
Atlantic Water	AW	>3.0	>34.65	Warm, saline water advected from the West Spitsbergen Current.
Transformed Atlantic Water	TAW	1.0–3.0	>34.65	AW that has been modified by cooling and mixing on the shelf.
Arctic Water	ArW	-1.5–1.0	34.30–34.80	Cold, relatively fresher water from the Arctic coastal current.
Intermediate Water	IW	>1.0	34.00–34.65	Formed by the vertical mixing of surface water with underlying AW/TAW.
Surface Water	SW	>1.0	<34.00	Locally formed by glacial melt and river runoff.
Winter Cooled Water	WCW	<-0.5	34.40–35.00	Dense water formed by winter cooling and brine rejection during sea ice formation.

Table 2. Water mass classification criteria for Kongsfjorden. *Note:* * – Water mass definitions follow Cottier *et al.* (2005).

An AI-Assisted Underwater Optical Monitoring System

System Design and Deployment

A custom-designed underwater optical monitoring system (IPR: 202441073989) was developed for this study in real-time, *in-situ* image acquisition and analysis (Fig. 2 and Fig. A1). The system consists of a 4K high-definition camera with a resolution of 3840×2160 pixels, housed within a robust, waterproof casing rated for polar conditions. To ensure consistent image quality in the variable and often low-light underwater environment of an Arctic fjord, the system is equipped with an adjustable LED lighting array with controllable intensity settings to optimize visibility under varying turbidity conditions. This array is calibrated to provide optimal illumination (500 to 1000 lumens) while minimizing backscatter and artificial visual artifacts

(Caimi *et al.* 2008).

At the core of the system is an onboard BeagleY-AI CPU, which performs real-time image processing and object detection directly in the field. The system was deployed by lowering it from the research vessel at each station. Vertical profiles were conducted by pausing the system at discrete depth intervals between 0 and 40 m to capture both video footage and static images. The monitoring system was typically maintained at each depth interval for approximately 3–5 minutes depending on visibility and fish activity. Due to the power constraints of the onboard battery system in cold water temperatures, each deployment session was limited to a maximum duration of two hours (Fig. A2).

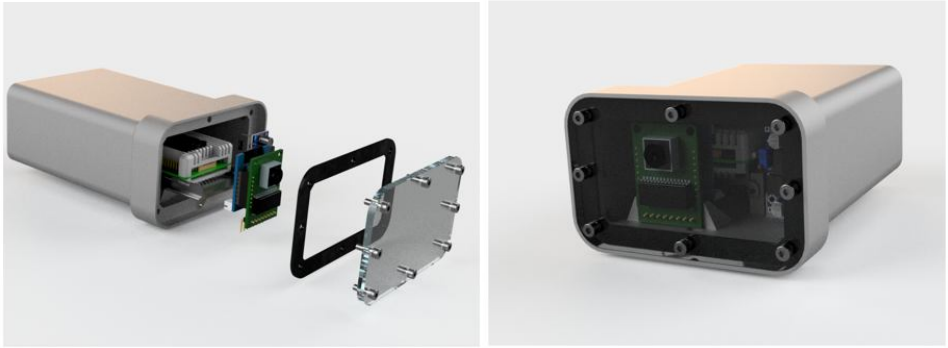


Fig. 2. The custom-built AI-assisted underwater monitoring system. The left panel shows an exploded view of the internal electronics, housing, and waterproof seal. The right panel shows the fully assembled unit ready for deployment.

Image Dataset Curation and Augmentation

A dataset was curated to train and validate the object detection model. The process involved several key steps:

- **Preprocessing-** Raw video footage was converted into individual image frames. Each frame was manually screened to discard images of poor quality, such as those with excessive motion blur, occlusions from debris, or severe lighting artifacts. To prevent model overfitting and reduce dataset redundancy, consecutive frames depicting identical scenes were identified and removed.

- **Image Enhancement-** To counteract the degrading effects of the underwater environment, such as light absorption and scattering that cause colour distortion and reduced contrast, a suite of image enhancement algorithms were applied. These included contrast and brightness normalization and background noise reduction, consistent with best practices in underwater imaging studies (Caimi et al. 2008).

- **Annotation-** A high-quality ground-truth dataset was created through manual annotation. Using the open-source software Labelling, bounding boxes were drawn around each visible fish specimen in the pre-processed images. Each bound-

ing box was assigned a species-specific class label: *G. morhua* or *B. saida*. The annotations were saved in the YOLO format, generating a text file for each image containing the class ID and normalized bounding box coordinates.

- **Data Augmentation-** To increase the size and diversity of the training dataset, thereby improving the model's generalization capabilities, a comprehensive data augmentation strategy was employed. This strategy was designed to simulate the wide range of conditions encountered in the Arctic marine environment and are given below.

- **Geometric Transformations-** Random transformations including rotation (± 15 degrees), horizontal flipping, scaling ($\pm 20\%$), and translation ($\pm 10\%$ of image height and width) were applied to account for variability in fish orientation, size, and position within the camera's field of view (Lin et al. 2001).

- **Photometric Transformations-** Random adjustments to brightness ($\pm 25\%$), contrast ($\pm 25\%$), and hue-saturation-value (HSV) colour space were applied to simulate the effects of changing ambient light levels, water depth, and time of day.

○ Environmental Simulation-To mimic the challenging optical properties of glacial fjord water, augmentations such as Gaussian blur and additive Gaussian noise were introduced. These techniques help

the model become more robust to images degraded by high turbidity and sensor noise, which are common in the study area (Yuan et al. 2022).

YOLOv8 Object Detection Framework

The object detection module was built upon the YOLOv8 (You Only Look Once, version 8) architecture, a state-of-the-art, single-stage object detection model known

for its exceptional balance of speed and accuracy (Qin et al. 2024). The architecture comprises three main components (Fig. 3).

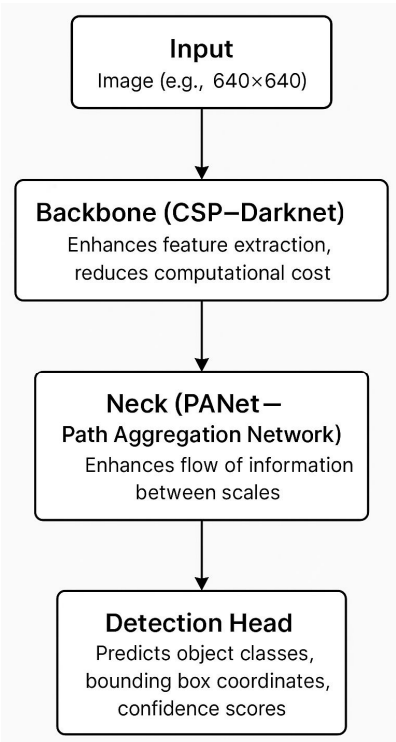


Fig. 3. The architecture of the YOLOv8-based deep learning model used for fish detection and classification. The model processes an input image through three main stages including (1) a Backbone for feature extraction, (2) a Neck for feature fusion, and (3) a Detection Head for final prediction.

Backbone (CSP-Darknet) A Cross Stage Partial Darknet (CSP-Darknet) serves as the backbone for efficient feature extraction. It utilizes a series of convolutional layers, batch normalization, and

Leaky ReLU activation functions to extract hierarchical features from the input image while minimizing computational redundancy.

Neck (PANet) A Path Aggregation Network (PANet) is employed as the neck of the model. It enhances the flow of information between feature maps of different scales, effectively fusing features from both lower and higher levels of the backbone. This multi-scale feature fusion is particularly crucial for detecting objects of varying sizes, a common challenge in underwater fish surveys (Xiao et al. 2024).

Detection Head YOLOv8 utilizes an anchor-free detection head, which simplifies the model by directly predicting the centre of an object and its dimensions rather than relying on predefined anchor boxes. This approach improves the model's flexibility and performance, especially for detecting small or irregularly shaped objects. The head outputs predictions for object class, bounding box coordinates, and a confidence score. Non-Maximum Suppression (NMS) is applied post-processing to eliminate redundant, overlapping bounding boxes for the same object.

The model was trained by optimizing a composite loss function, L_{total} , which is a weighted sum of three components:

$$L_{total} = \lambda_{box}L_{box} + \lambda_{obj}L_{obj} + \lambda_{cls}L_{cls}$$

where L_{box} is the bounding box regression loss (calculated using Complete IoU, or CIoU, to account for overlap, centre point distance, and aspect ratio), L_{obj} is the objectness loss (a measure of confidence that an object is present in each box), and L_{cls} is the classification loss (which penalizes incorrect species predictions). Both L_{obj} and L_{cls} are calculated using Binary Cross-Entropy (BCE) loss.

Training was conducted using an empirically optimized set of hyperparameters selected through iterative validation experiments to maximize detection accuracy while minimizing overfitting. Input images were resized to 640×640 pixels. The model was trained with a batch size of 16 for 250 epochs using the AdamW optimizer with an initial learning rate of 0.001. The weights for the loss function components (λ) were set to $\lambda_{box}=8.0$, $\lambda_{obj}=1.0$, and $\lambda_{cls}=0.5$. This differential weighting places a higher emphasis on accurate localization (bounding box regression), which is critical for distinguishing between morphologically similar fish in challenging, low-visibility underwater environments.

Model Validation and Performance Metrics

The curated and augmented dataset was partitioned into training (70%), validation (20%), and testing (10%) subsets. The validation set was used during training to monitor performance and prevent overfitting, while the independent test set was used for the final evaluation of the model's performance. The model's accuracy was quantified using several standard metrics (TP- True Positive, FP- False Positive, FN- False Negative) as given below.

Precision- The fraction of correct positive predictions among all positive predictions ($TP/(TP+FP)$). **Recall-** The fraction

of actual positives that were correctly identified ($TP/(TP+FN)$).

F1-Score- The harmonic means of precision and recall ($2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$).

mean Average Precision (mAP)- The primary metric for object detection, representing the average of the Average Precision (AP) over all classes. We report mAP at an Intersection over Union (IoU) threshold of 0.50 (mAP50) and as an average over IoU thresholds from 0.50 to 0.95 (mAP50-95).

Results

AI-Based Species Detection Model

The performance of the YOLOv8 for species classification was validated, with the results visualized in a confusion matrix (Fig. 4). The model demonstrated a high overall accuracy of 86.3% in differentiating between Polar cod (*B. saida*) and Atlantic cod (*G. morhua*) from *in-situ* imagery. Specifically, the model correctly identified 109 of the 132 observed Polar cod (82.6% sensitivity) and 124 of the 138 observed Atlantic cod (89.9% sensitivity).

Misclassifications were relatively low, with 23 Polar cod incorrectly identified as Atlantic cod and 14 Atlantic cod misidentified as Polar cod. This robust performance underscores the reliability of the AI-driven approach for accurately quantifying the presence and abundance of these two key species, thereby providing a solid foundation for the ecological conclusions drawn in this study.

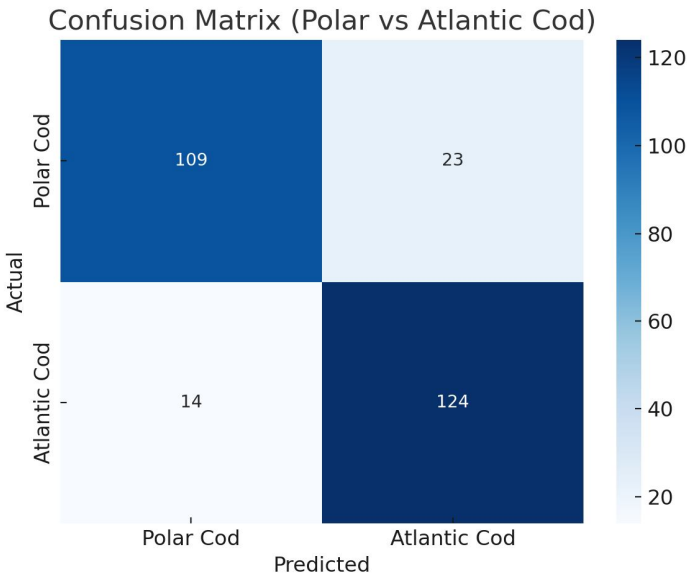


Fig. 4. Confusion matrix of classification performance of the AI model, YOLOv8. The matrix displays the number of correct (diagonal values) and incorrect predictions for Polar and Atlantic cods based on the validation dataset.

The YOLOv8 model, trained on the curated dataset of underwater images from Kongsfjorden, demonstrated high performance and reliability in detecting and

classifying the two target cod species (see Fig. 5 and Fig. 6). The quantitative performance metrics, evaluated on the independent test set, are summarized in Table 3.

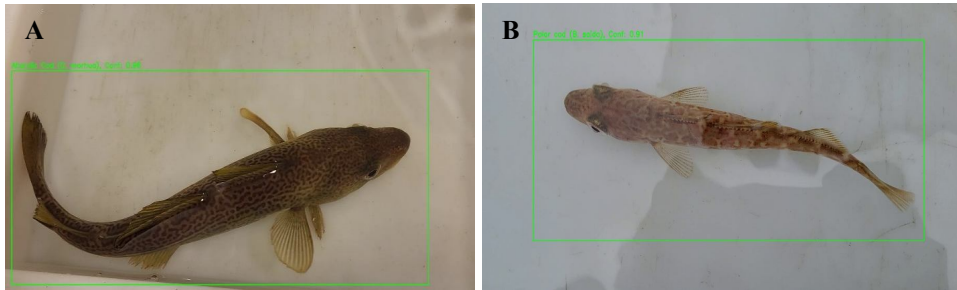


Fig. 5. Performance of the YOLOv8 on images of captured specimens used for training and validation. The system correctly identifies (A) an Atlantic cod (*G. morhua*) and (B) a Polar cod (*B. saida*) in a controlled environment.

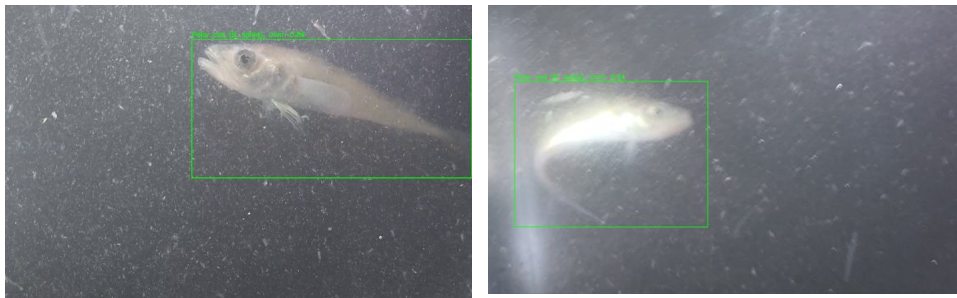


Fig. 6. *In-situ* application of the AI monitoring system in Kongsfjorden. The images show real-time, successful identification of Polar cod (*B. saida*) under typical survey conditions, characterized by high turbidity.

Class	Precision	Recall	F1 Score	Accuracy
Polar cod (<i>B. saida</i>)	0.88	0.83	0.85	0.92
Atlantic cod (<i>G. morhua</i>)	0.91	0.87	0.89	0.94
Confusion Matrix	Predicted Atlantic Cod	Predicted Polar Cod		
Actual Atlantic Cod	True Positive: 87%	False Negative: 13%		
Actual Polar Cod	False Positive: 12%	True Positive: 88%		

Table 3. Performance Metrics and Confusion Matrix of the YOLOv8 Detection Model.

The model achieved high precision and recall for both species, indicating a low rate of both false positives and false negatives. For Atlantic cod, the precision was 0.91 and recall was 0.87, resulting in an F1-Score of 0.89. For the native Polar cod, the model achieved a precision of 0.88 and a recall of 0.83, with an F1-Score of 0.85. The overall accuracy for both classes ex-

ceeded 92%, confirming the robustness of the model for this specific application. The confusion matrix further illustrated the strong discriminative power of the model, with the vast majority of individuals correctly classified and minimal confusion between the two morphologically similar species.

Distribution of Cod Species

The deployment of the AI-assisted optical monitoring system revealed an unambiguous pattern in the distribution of Atlantic cod. An abundance of *G. morhua* was detected exclusively during the 24 September 2024 survey, and these detections were concentrated at the innermost monitoring station, KG1. During this single

monitoring event, the system identified more than 20 Atlantic cod individuals, suggesting a significant influx of the species into the inner fjord basin. In contrast, during the other monitoring periods of May-June 2023, detections of Atlantic cod were negligible or entirely absent across all stations.

Hydrography and Intrusion of Atlantic Water

The hydrographic conditions in Kongsfjorden exhibited significant variability during the monitoring periods, reflecting the dynamic interplay of Atlantic, Arctic, and freshwater influences. Temperature-Salinity (T-S) diagrams (Fig. 7) illustrate the distinct water mass compositions during the surveys in 2023 and 2024. In the early summer of 2023, cooler and fresher surface water mass overlying deeper, more saline water masses were in seen in Kongsfjorden indicating a stratified water column. However, AW presence was weak from ~ 40-90 m depth in June 2023 with a maximum temperature of ~ 5°C (Fig. 4). TAW was found below it. During this period, the upper water column (0-40 m), where biological monitoring was focused, was predominantly composed of Surface Water mass (SW) and Intermediate Water (IW).

On the contrary, the late summer/early autumn period of 2024 was marked by a strong presence of warm, saline AW. On September 24, 2024, the T-S profile revealed that AW dominated a substantial portion of the water column, extending from 50 to 250 m depth. The maximum temperature of AW in 2024 was ~8.0 °C, almost 3°C warmer than that in June 2023. This warmer and intense AW presence fundamentally altered the hydrography of the entire fjord. The Intermediate Water (IW) layer that had a maximum temperature of ~4.5°C in June 2023 also showed elevated temperature of ~7.5°C in 2024. This indicated a strong vertical mixing of SW with the warmer AW layer beneath it leading to the formation of warmer IW. This hydrographic state in 2024 represents a period of intense Atlantification of Kongsfjorden system.

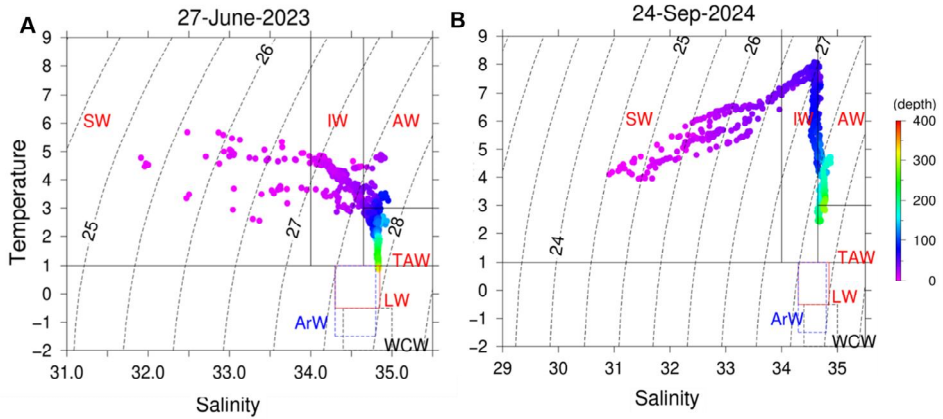


Fig. 7. Temperature-Salinity (T-S) diagrams from CTD casts in Kongsfjorden during surveys in (A) 2023 and (B) 2024. The x-axis represents Salinity (PSU), and the y-axis represents Temperature (°C). Water masses are defined following Cottier et al. (2005), and the colour scale indicates water depth.

The observed biological observation aligns directly with the Kongsfjorden hydrography. The mass presence of Atlantic cod in the upper waters of the inner fjord coincided precisely with the period of the intense intrusion of warm Atlantic Water into the fjord in 2024. The near absence of Atlantic cod during 2023, characterized by weaker AW influence and more stratified, colder surface conditions compared to those observed during September 2024, es-

tablishes a strong temporal correlation between the oceanic condition and the biological response. Table 4 provides a summary that directly juxtaposes the fish detection data with the prevailing hydrographic conditions for each survey period, highlighting the relationship. During all survey periods, only zooplankton, invertebrates, and sculpins were consistently observed, with cod species being a notable variable.

Survey Months	Station	<i>G. morhua</i> detections	<i>B. saida</i> detections	Dominant Water Mass (0-40m)	Max Temperature (°C)
May-June 2023	KG1-KG4	0	Low	Surface Water (SW)	3.5
	KG1-KG4	< 5	Low	Intermediate Water (IW)	4.8
September 2024	KG1	High Abundance	Low	Intermediate Water (IW)	7.1
	KG2-KG4	Low	Low	Intermediate Water (IW)	6.5

Table 4. Summary of fish detections and corresponding hydrographic conditions.

Discussion

Borealization driven by Atlantic Water inflow

The results of this study provide compelling evidence for the mechanistic link between biological community change and oceanographic forcing in an Arctic fjord. The contrast between the abundance of Atlantic cod observed exclusively during the intense AW intrusion of September 2024 that occupied maximum fjord depth and the near-total absence of Atlantic cod in 2023 constitutes a direct observation of event-driven borealization. While long-term warming trends set the stage for such invasions, our findings suggest that the actual penetration of boreal species into sensitive inner-fjord habitats is not a continuous, gradual process. Instead, it appears to be facilitated by acute, large-scale advective events that temporarily transform the local environment, making it thermally suitable for boreal fish invaders, particularly Atlantic cod (*G. morhua*) (Cottier *et al.* 2005). Similar Atlantification-driven expansions of boreal fish species into Arctic marine ecosystems have also been reported from the Barents Sea and other regions around Svalbard, where warming Atlantic Water inflow has altered species distributions and community structure (Fossheim *et al.* 2015, Descamps *et al.* 2026). The detection of Atlantic cod in the upper layer identified as IW, is significant. The formation of warmer IW through the vertical mixing of surface waters with the underlying warmer AW provides a direct physical pathway, demonstrating how changes in

the deep-water masses can impact the near-surface ecosystem where most biological activity occurs.

This observation strongly supports the concept of Kongsfjorden as a harbinger of future Arctic conditions (Bischof *et al.* 2019). Under the ongoing circumstance of the well-documented transition of Kongsfjorden towards a more Atlantic-dominated hydrographic regime (Polyakov *et al.* 2017) our study captures a snapshot of this transition in action, illustrating how increasing frequency or intensity of AW intrusions, a predicted consequence of climate change (Asbjørnsen *et al.* 2020), would likely accelerate the establishment of boreal species and the restructuring of native Arctic communities (Dischereit *et al.* 2024).

The successful implementation and validation of our deep learning model, which achieved 86.3% accuracy, is a significant methodological advancement for monitoring dynamic Arctic ecosystems. This provides strong confidence in our reported observations of Atlantic cod incursions (Polyakov *et al.* 2017). Nevertheless, the error rate of the model, though minimal, underscores inherent challenges of automated species identification *in situ*. Addressing these challenges through expanded training datasets represents a critical next step in developing even more powerful tools for tracking the long-term impacts of climate change and borealization.

Ecological Implications

The establishment of a persistent Atlantic cod population in Kongsfjorden carries profound ecological implications, foremost among them being the threat to the native polar cod. As a larger, faster-growing, and more aggressive species, Atlantic cod is a superior competitor and a potential

predator of polar cod (Kunz *et al.* 2016, Szczucka *et al.* 2017). The two species have differing thermal optima and polar cod are highly adapted to sub-zero temperatures (preferred range -1.5 to 5.0°C). Atlantic cod thrive in warmer conditions (preferred range 0–12°C) that are becom-

ing increasingly common in the fjord (Drinkwater 2005, Thorsteinson and Love 2016). As the fjord continues to warm, the environmental conditions will progressively favour the boreal invader at the expense of the endemic Arctic species (Dahlke et al. 2020).

The increased abundance of fish, such as cod (*G. morhua*) and haddock (*Melanogrammus aeglefinus*), has been reported near the western coast of Spitsbergen and entering fjords together with Atlantic waters. The presence of Atlantic cods in the deeper outer regions of the two west Spitsbergen fjords, a relatively cooler fjord- Hornsund and warmer Kongsfjorden, in 2014 were explained by the intense inflow of Atlantic water on the shelf (Szczucka et al. 2017). Isfjorden which is another west Svalbard fjord has also shown Atlantic cod presence that was owed to the increase in Atlantic water intrusions to the fjord. The hydroacoustic data collected in 2022 further suggests

change in the food web and a decline in native polar cod in Kongsfjorden confirming that the process of establishment of non-Arctic species Atlantic cod in Kongsfjorden continues (Gorska et al. 2023). Studies from other rapidly warming Arctic regions, such as the Barents Sea, have already documented a northward retreat of polar cod populations coinciding with the expansion of Atlantic cod and other boreal species (Fossheim et al. 2015, Huserbråten et al. 2015). The high abundance of Atlantic cod observed in the inner Kongsfjorden in the present study, a region that may serve as a critical refuge and nursery habitat for polar cod, suggests that this competition will occur in the most sensitive areas of the ecosystem. The displacement of polar cod would represent the loss of not just a species, but a key functional component of the Arctic food web, with consequences that would reverberate through the entire ecosystem.

Implications to classic food web

The introduction of a novel apex predator into a relatively simple, established food web can trigger a trophic cascade, where impacts ripple down through multiple trophic levels, fundamentally restructuring the ecosystem (Frank et al. 2005). The establishment of Atlantic cod in Kongsfjorden, an ecosystem historically structured around the smaller, mid-trophic level polar cod, presents a classic scenario for such a cascade. The observed changes may represent a shift beyond "ecosystem maturation," a process where new species are added, increasing diversity and complexity (Węśławski et al. 2017). The introduction of increasingly dominant predator is not merely an addition but a functional replacement at the top of the fish food web, which is more characteristic of a fundamental regime shift.

Increased predation by and resource competition with Atlantic cod leads to a

significant decline in the population of native polar cod. This has been observed in other formerly cod-dominated ecosystems where the removal of the top predator led to dramatic increases in their prey (Frank et al. 2005). Polar cod are a primary consumer of large, lipid-rich Arctic zooplankton, such as *Calanus* copepods and ice-associated amphipods (Hop and Gjøsæter 2013). A reduction in polar cod abundance could release these zooplankton populations from their main fish predator, potentially leading to an increase in their biomass. Changes in the grazing pressure exerted by large zooplankton could, in turn, alter the phytoplankton community structure, potentially favouring smaller, less palatable species and affecting primary production dynamics (Ardayna and Arrigo 2020). The most severe consequences would likely be felt at higher trophic levels. Many Arctic marine mam-

mals (*e.g.*, ringed seals, beluga whales) and seabirds are highly dependent on the dense, energy-rich schools of polar cod (Welch *et al.* 1992, Geoffroy *et al.* 2023). The replacement of this crucial prey base with a more dispersed and less accessible

predator like Atlantic cod could lead to reduced foraging success, poor body condition, and ultimately, population declines for these iconic Arctic species (Florko *et al.* 2021).

Innovations in Arctic Monitoring

This study underscores the transformative potential of integrating AI with optical technologies for Arctic biomonitoring. The developed system offers a non-invasive, scalable, and potentially cost-effective method for collecting high-frequency, species-specific data on fish abundance and distribution (Philip *et al.* 2023). This approach overcomes many of the inherent limitations of traditional net-based surveys, which are labour-intensive, destructive to fragile benthic habitats, and provide only spatially and temporally limited data points. In a rapidly changing environment where ecosystem dynamics can shift on

timescales of weeks or even days, the ability to deploy autonomous systems for continuous observation is paramount.

The validation of this technology provides a powerful new tool for enhancing vital long-term monitoring programs in the Arctic. Integrating automated, AI-driven optical sensors into these existing observational networks in Kongsfjorden and other Arctic region could dramatically increase the resolution and scope of biological data collection, enabling a more comprehensive and timely assessment of ecosystem health and response to environmental perturbations.

Limitations and Future Research Horizons

This study, while providing a clear and compelling snapshot of an ecological shift, has several limitations that must be acknowledged. First, while the AI model demonstrated high accuracy, the potential for misidentification, particularly in conditions of high turbidity or with partially occluded individuals, cannot be eliminated. Moreover, continuous model refinement with larger and more diverse datasets is necessary. Second, the temporal coverage of our sampling, is composed of discrete campaigns. This approach may miss finer-scale seasonal or short-term variations in species presence and hydrographic conditions. Third, the fixed field-of-view of a stationary camera system can introduce bias in abundance estimates, as it does not account for fish behaviour, such as attraction or avoidance, and cannot capture the full range of activity patterns.

Future research should aim to address these limitations and build upon the findings of this study. A key priority is the development of systems capable of year-round, autonomous deployment to capture the full annual cycle of both oceanographic and biological dynamics, including the critical but under-sampled polar night period. Expanding the monitoring network to include more stations, both within Kongsfjorden and in other Svalbard fjords, would provide a more comprehensive understanding of the spatial scale of borealization. Finally, the most powerful approach will involve the integration of multiple monitoring technologies. Combining AI-driven optical data with hydro-acoustic surveys for biomass estimation, eDNA metabarcoding for comprehensive biodiversity assessment, including cryptic species, and traditional biological sam-

pling for physiological and trophic analysis will create a multi-faceted, ecosystem-wide monitoring framework capable of

meeting the challenges of a rapidly changing Arctic.

Conclusion

This research provides an unambiguous link between a significant biological response, the mass influx of a boreal predator into a high-Arctic fjord, and a strong Atlantic water intrusion. The findings underscore a critical mechanism of climate-driven borealization, demonstrating that long-term warming trends can manifest as rapid, pulsed invasions that have the potential to accelerate ecosystem reorganization. The successful deployment and validation of a custom-made, AI-driven camera system offers a scalable and non-invasive tool for high-resolution biomonitoring, representing a significant technological advancement for research in remote and fragile polar environments.

The observed replacement of the native, ice-associated polar cod by the inva-

sive Atlantic cod marks a significant ecological shift in Kongsfjorden. This transition could have profound and cascading consequences for the entire Arctic marine food web, potentially destabilizing trophic interactions and impacting the predators that rely on the unique, energy-rich native fauna. Given the accelerating rate of climate-induced changes and the complex dynamics of borealization, the need for continued, interdisciplinary, and high-resolution monitoring is more urgent than ever. Future research must focus on the long-term implications of this species turnover for trophic dynamics, ecosystem stability, and the resilience of native Arctic communities in the face of persistent environmental change.

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Supplementary material

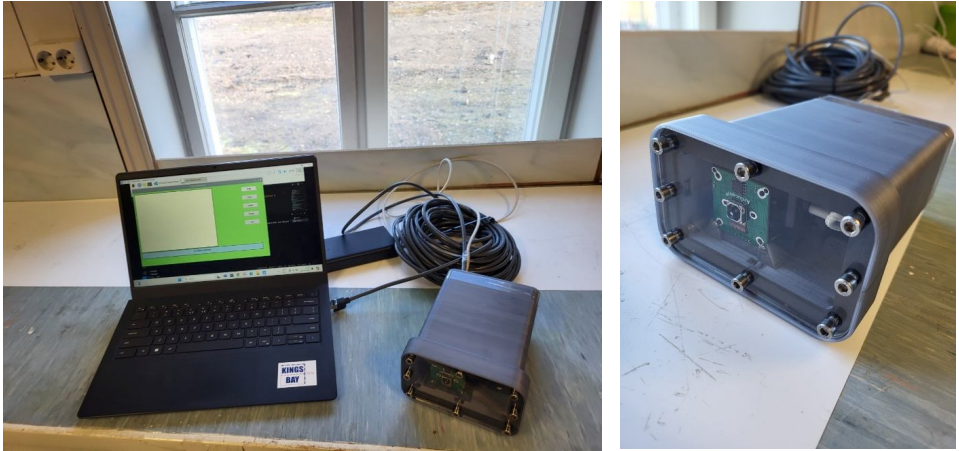


Fig. A1. A custom-designed underwater optical monitoring system (IPR: 202441073989) was developed for real-time, in-situ image acquisition and analysis.

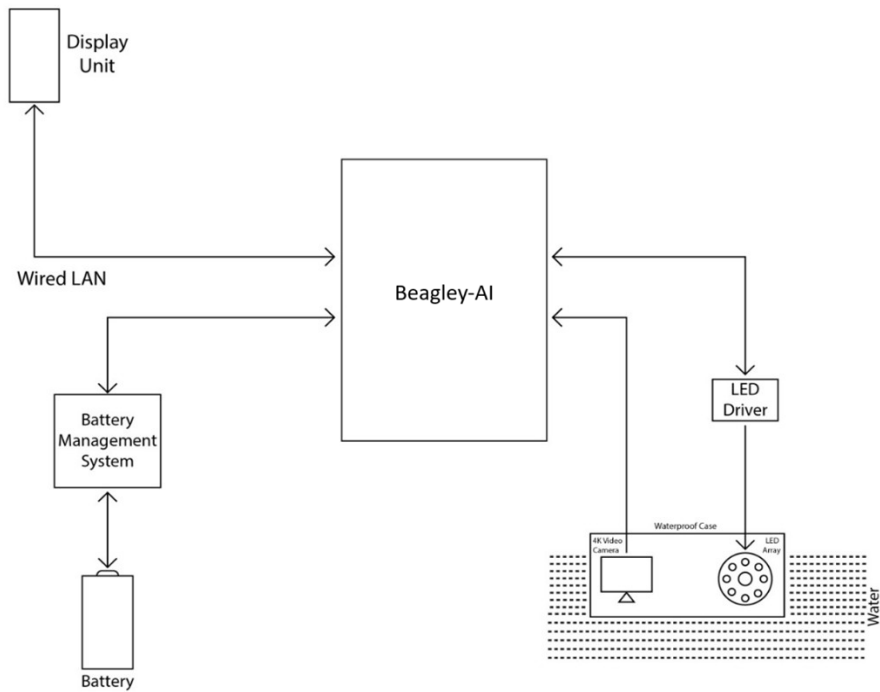


Fig. A2. Illustration of custom-designed underwater optical monitoring system design architecture.